Riccardo **Buscicchio** | Curriculum Vitæ

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Academic positions

Gran Sasso Science Institute **L'Aquila, Italy**
Tenure-track assistant professor *2026 - 2032*

Università degli Studi di Milano-Bicocca **Milan, Italy**
Postdoc (Assegnista di ricerca), Department of Physics "G.Occhialini" *2021 - 2024*

- *Main activity*: development of LISA data analysis ground-segment for the Italian Space Agency (Phase A).

Università degli Studi di Milano-Bicocca **Milan, Italy**
Postdoc (Assegnista di ricerca), Department of Physics "G.Occhialini" *2024 - current*

- *Main activity*: development of LISA data analysis ground-segment for the Italian Space Agency (Phase B).

Education

University of Birmingham **Birmingham, UK**
Ph.D., School of Physics & Astronomy *2017-13/07/2022*

- *Supervisor*: A. Vecchio. Thesis resulted in 6 short-author publications.
- *Thesis Title*: Topics in Bayesian population inference for Gravitational Wave Astronomy

Università degli Studi di Pisa **Pisa, Italy**
Master's degree in Theoretical physics *2013-2016*

- *Supervisor*: G. Cella. Thesis resulted in one short-author publications.
- *Thesis title*: An improved detector for non-gaussian stochastic background of gravitational waves.

Columbia University **New York NY, USA**
INFN-NSF Summer Internship *Jun-Sept 2013*

- *Supervisor*: S. Marka, I. Bartos.

Metrics

Publications:

38 short-author papers published in major peer-reviewed journals
(out of which **8** first-authored papers and **11** lead by supervised student).
14 collaboration papers with substantial contribution, published in major peer-reviewed journals
60 collaboration papers in total, published in major peer-reviewed journals
5 papers in submission stage,
2 other publications (thesis, white papers, reviews)

Total number of citations: >20000. **h-index**: 28 (from ADS and iNSPIRE record).
Web links to list services: ADS; iNSPIRE; arXiv; orcid.
Full list of publications available at `www.riccardobuscicchio.com/publications`.
Full list of presentations available at `www.riccardobuscicchio.com/talks`.

Codes & Datasets

Title	Code	Dataset	Zenodo DOI	Public
○ Bahamas Galactic Foreground	✓	✓	10.5281/zenodo.17527235	✓
○ Bahamas	✓		10.5281/zenodo.16087705	✓
○ Hypertriangulation Map	✓		10.5281/zenodo.13897708	✓
○ Bayesian PowerLaw Sensitivity	✓	✓	10.5281/zenodo.14384633	✓
○ Milky Way Satellites		✓	10.5281/zenodo.3668904	✓
○ LISA stellar BBH catalogues and samples	✓	✓	10.5281/zenodo.14426778	✓
○ LISA MBHB catalogues and samples		✓	10.5281/zenodo.13787674	✓
○ Chirp Gravitational Wave Alerts	✓		10.5281/zenodo.3525063	✓

Grants, Prizes, & Awards

Career prizes:

- ○ Braccini PhD Thesis Prize, Gravitational Wave International Committee honorable mention. *2021*
- ○ Michael Penston PhD Thesis Prize, Royal Astronomical Society runner-up prize. *2021*

Grants:

- ○ Royal Society of Edinburgh "Gravity under the lens" (coI, PI Dr. Berry, Univ. of Glasgow, UK) *2025-2027*
- ○ EuroHPC PRACE "LABELFIT" proposal, 224k CPUh on Leonardo DCGP *2025*
- ○ EuroHPC PRACE "LISA-FIT" proposal, 100k CPUh on Leonardo BOOSTER *2023*
- ○ Google Cloud for Researchers, 4kEUR Google Cloud Research Credits *2023*
- ○ CINECA ISCRA Type-C project "LISA-MW" proposal, 10k CPUh at the Italian National HPC center. *2022*
- ○ Travel Grant, Horizon 2020 AHEAD 2020 (High Energy Astrophysics) *2021*
- ○ Travel Grant, American Physical Society, DGRAV Student Travel Grant *2020*
- ○ Travel Grant, Institute of Physics Student Travel fund *2019*
- ○ Travel Grant, Royal Astronomical Society, UK. *2018*

Invited research visits:

- ○ Kavli visitor program research visit — DAMTP Univ. Cambridge, Cambridge, UK *07/10/2024-18/10/2024*
- ○ Research visit — Max Planck Institute for Astrophysics, Garching, DE *23/11/2023-30/11/2023*
- ○ Research visit — Albert Einstein Institute, Potsdam, DE *16/04/2023-23/04/2023*
- ○ Research visit — Univ. of Stavanger, Stavanger, NO *08/02/2023-29/03/2023*
- ○ AHEAD 2020 program research visit — APC, Univ. Paris Diderot, Paris, FR *01/12/2023-07/01/2023*
- ○ Research visit — APC, Univ. Paris Diderot, Paris, FR *26/10/2022-30/11/2022*
- ○ Research visit — Univ. of Stavanger, Stavanger, NO *26/09/2022-19/10/2022*
- ○ Research visit — Univ. of Birmingham, Birmingham, UK *05/07/2022-10/07/2022*
- ○ Research visit — Columbia University, New York, US *06/04/2022-13/04/2022*

Student supervision

According to current national regulations, as a research fellow I cannot be officially appointed as supervisor of students at any level. However, upon agreement with the relevant permanent staff, I have supervised the work of students in the percentages shown below.

Ph.D. student co-supervisor:

- ○ B. Saiguhan, University of Milano-Bicocca, PhD, 100% *2025-2028*
- ○ L. Viganò, University of Milano-Bicocca, PhD, 80% *2025-2028*
- ○ F. Nobili, University of Insubria, PhD, 100% *2024-2027*
- ○ A. Spadaro, University of Milano-Bicocca, PhD, 100% *2022-2025*
- ○ F. Pozzoli, University of Insubria, PhD, 100% *2022-2025*

MSc student co-supervision:

- ○ R. Bosoni De Martini, University of Milano-Bicocca, Master's thesis, 100% *2025-2026*
- ○ L. Viganò, University of Milano-Bicocca, Master's thesis, 100% *2024-2025*
  (now PhD student at Univ. of Milano-Bicocca)
- ○ M. Bellotti, University of Milano-Bicocca, Master's thesis, 100% *2024-2025*

(now PhD student at Observatoire Cote d'Azur)
- D. Chirico, University of Milano-Bicocca, Master's thesis, 100% *2023-2024*
- S. Corbo, Politecnico di Milano, Master's thesis, 100% *2023-2024*
- R. Rosso, University of Pisa, Master's thesis, 80% *2023-2024*
(now PhD student at Univ. of Pisa)
- G. Astorino, University of Pisa, Master's thesis, 80% *2023-2024*
(now PhD student at Univ. of Pisa)
- M. Piarulli, University of Milano-Bicocca, Master's thesis, 100% *2022-2023*
(now PhD student at Univ. of Toulouse)
- A. Spadaro, University of Milano-Bicocca, Master's thesis, 100% *2021-2022*
(now PhD student at Univ. of Milano-Bicocca)
- A. Carzaniga, University of Milano-Bicocca, Master's thesis, 100% *2021-2022*
- A. Geminardi, University of Milano-Bicocca, Master's thesis, 100% *2021-2022*
(now PhD student at Univ. of Pavia)
- E. Finch, University of Birmingham, Year 4 project, 50% *2018*
- V. Spasova, University of Birmingham, Year 4 project, 50% *2018*

BSc student supervision:
- H. P. G. Carabajo, University of Milano-Bicocca, Bachelor's thesis, 100% *2023-2024*

Taught classes ***Acad.Year***
- **Inference models in LISA**, PhD course, Univ. of Trento (Italy). *2026*
- **Current and future challenges in GW astronomy**, PhD course, Milano-Bicocca (Italy). *2023*
Content: Introduction to gravitational-wave data analysis.
Space-based and ground-based interferometers.
Source populations: compact object binaries, stochastic backgrounds.
- **Mathematical physics and gravity** (MAF900), PhD course, Univ. of Stavanger (Norway). *2023*
Content: Introduction to gravitational-wave data analysis. Space-based interferometers.
Source populations: compact object binaries, stochastic backgrounds.
Signal modeling: galactic and extragalactic compact object binaries
astrophysical and cosmological stochastic backgrounds.
Signal detection and parameter estimation: frequentist and Bayesian approaches.
Advanced stochastic sampling techniques.
- **Lecture in Astrostatistics** (F5802Q014/20), Astrophysics MSc, Univ. of Milano-Bicocca (Italy) *2022*
Content: Source populations: compact object binaries, stochastic backgrounds.
Introduction to gravitational-wave data analysis. Space-based interferometers.
Signal detection and parameter estimation: frequentist and Bayesian approaches.

Tutoring ***Acad.Year***
- **Python Computing Lab**, Bachelor's degree in Physics, Univ. of Birmingham, UK *2017-2021*
Content: Python programming, simulation of physical systems in celestial
and classical mechanics, thermodynamics, electromagnetism. Data analysis.
- **Maths for physicists**, Bachelor's degree in Physics, Univ. of Birmingham, UK *2017-2019*
Content: Linear algebra, differential and integral calculus, differenzial equations
Groups and representations theory
- **Physics and communication skills**, Master's degree in Physics, Univ. of Birmingham, UK. *2019*
Content: LaTeX basics. Drafting of lab reports, publications, seminars

Academic service, editorial and research responsibilities

International collaboration responsibilities
- LISA Consortium AstroWG coordinator, project "Updatable predictions for stBBH in LISA" *2025-2026*
- PAQA co-chair of the Coordination Unit L2D (Global Fit, ESA LISA Project Office) *2024-2026*
- DDPC-Italy Lead of the LISA DDPC national coordination initiative *2024-2026*

Journal referee

- Physical Review Letters
- The Astrophysical Journal Letters
- The Open Journal of Astrophysics
- NASA Technology Transfer Program
- IoP Classical and Quantum Gravity
- Institute of Physics Trusted Reviewer Excellence program
- Physical Review D
- Physics Letters B
- Journal of Cosmology and Astroparticle Physics
- Monthly Notices of the Royal Astronomical Society
- Institute of Physics Machine Learning: Science and Technology

Editorial responsibilities

- Editorial board Lensing working group in the LIGO,Virgo, KAGRA collaboration *2023*
- Co-editor of a Living Review in relativity issue on "LISA data analysis" *2022-2023*

Conference organizer

- *LISA Without Frontiers* Workshop Sprint Session Coach, Sexten, IT. *2026*
- *LISA Distributed Data Processing Center June Workshop*, Milan, IT. *2025*
- IFPU focus week on "*Emerging methods in GW population inference*", Trieste, Italy. *2024*
- *LISA Astrophysics Working Group Conference*, Birmingham, UK. *2022*
- *Gravitational-wave populations: what's next?*, Milan, Italy. *2023*
- *Gravitational-wave Excellence Alliance Training (GrEAT) PhD school*, Birmingham, UK. *2019*
- *Gravitational-wave Open Science Center First Open Data Workshop*, (online) *2019*

Professional recognition and service

- National Scientific Habilitation (ASN) as associated professor. *2023*
  in Astrophysics, Cosmology and Space Science (GSD 02/PHYS-05 - PHYS-05/A)
- French Qualification for teaching in Higher Education in Astrophysics (Sec.34), *2023*
  French Ministry of higher education and research (qualification no.23234388826).
- LSC Academic Advisory Committee. *2019-2021*

Memberships

- LISA Distributed Data Processing Center, full member. *since 2024*
- LISA Consortium, core member. *since 2018*
- Italian Center for Supercomputing (ICSC). *since 2021*
- TEONGRAV National Initiative (Gravity Theory) *since 2021*
  Italian National Institute for Nuclear Physics (INFN).
- LIGO, Virgo, Kagra Collaboration, full member. *since 2017*
- Italian Society of General Relativity and Gravitational Physics (SIGRAV) *since 2021*
- Istituto Nazionale di Astrofisica (INAF) *since 2021*
- American Physical Society (APS), member.
- Italian Physical Society (SIF), member *2021*
- Royal Astronomical Society (RAS), fellow. *2018-2021*

Date: October 14, 2026